

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 22

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Orientation: south-facing windows (northern hemisphere) maximise passive solar gains; rooms needing warmth on the sunny side; fewer / smaller windows on the north side where losses are greatest; solar screens / brise soleil prevent summer overheating • Low SA:V (compact shapes, terraced / joined buildings, fewer storeys of small footprint) means less external surface per unit volume through which heat is lost • Earth sheltering: the ground is warmer than winter air and shields the building from wind, so conduction and wind-chill losses are reduced	6
01.2	• Cement manufacture (heating limestone) is very energy intensive so concrete has high embodied energy • Limecrete has about half the embodied energy of concrete; rammed earth, straw bales and lime mortar are lower still	2
01.3	• Warm stale air leaving and cold fresh air entering pass through the exchanger in opposite directions (counter-current) separated by thin conducting pipes / plates with a large surface area • Heat transfers from the outgoing to the incoming air so fresh air enters already warmed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-232; AQA spec 3.3.4.2.2.	
02.1	• LEO: high resolution and coverage of the whole Earth (including the poles) allows detailed mapping of ice area / thickness and forest cover • But each area is only revisited every 15 days so they cannot follow rapidly changing weather • Geostationary: constant view of the same region allows continuous monitoring of cloud / storm movements, though the poles are not visible and detail is lower	3

Question	Marking guidance	Marks
02.2	<i>Any 3 from:</i> • Wind velocity • Ocean currents / sea surface temperature • Wave height • Ice thickness • Vegetation cover • Sea level • Atmospheric temperature / composition	3
02.3	• Data for a past year (eg 1900) are fed in and the model's prediction for a later year (eg 2000) is compared with the real outcome • If the prediction matches, the model can be trusted (with caution); it is continually modified as more data are collected and differences analysed	2
02.4	• Models need data from all regions / oceans / altitudes; gaps (eg oceans, polar regions, past periods) introduce uncertainty into predictions • Unrepresentative data (eg urban heat islands, short time series) would bias the model; remote sensing helps fill gaps	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 102; AQA spec 3.2.1.2.	
03.1	• A waste is assumed to cause pollution until research shows it is unlikely to do so	1
03.2	• If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution • If polluters can escape their responsibilities / penalties are low, there is little incentive to comply	2
03.3	<i>Any 2 from:</i> • Marine discharges where currents dilute and disperse emissions • Emissions downwind of urban areas • Not discharging waste onto permeable rock above an aquifer	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
04.1	• At times of low demand / surplus supply (night, windy periods) surplus electricity pumps water from the lower to the upper reservoir, storing it as gravitational potential energy • At peaks in demand water flows down through the turbines generating electricity • Can go from standby to full power in under 15 seconds so it meets sudden peaks (TV pickup) that fossil / nuclear stations cannot follow	3
04.2	• Mass = $7 \times 10^6 \times 1000 = 7 \times 10^9$ kg • PE = $m g h = 7 \times 10^9 \times 9.8 \times 500$ • = 3.43×10^{13} J (accept 3.4×10^{13})	3
04.3	• The electricity used for pumping is surplus that would otherwise be wasted / lost as heat, or cheap off-peak electricity • It avoids building rapid-response low-efficiency peaking plants (open-cycle gas turbines) and allows intermittent renewables to be used	2

Question	Marking guidance	Marks
04.4	<i>Any 2 from:</i> • Two water bodies / sites with a large height difference close together (mountainous) • Impermeable, stable rock • Close to the grid / demand • Adequate water supply	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222; AQA spec 3.3.4.1.	
05.1	• Net held with its straight edge on the river bed facing the current (open to the flow) • The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) • Same substrate type, water depth and flow rate at both sites; same net mesh; repeat samples; invertebrates identified with a key and counted	3
05.2	<i>Any 3 from:</i> • Difficult to standardise the vigour / area kicked • Buried organisms or those fixed to stones are not dislodged • Some organisms swim and avoid the net • Flow rate varies and affects capture • Fragile organisms destroyed	3
05.3	• A fixed area of bed is enclosed by a frame so the sampled area is standardised; side nets reduce escapes • But only works in fast enough flow and is hard to set on rough substrates / deep water	2
05.4	• Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event • Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
06.1	• B	1
06.2	• TBT is highly toxic to marine molluscs / crustaceans and an endocrine disruptor altering growth and reproduction of oysters / whelks even at very low concentrations • It bioaccumulated in shellfish and could pass to humans eating them • It was assumed dilution would make it harmless (illustrates the need for the precautionary principle); now prohibited and replaced by copper	3
06.3	• In the low-oxygen conditions of the spoil heap iron is in its soluble reduced form and is leached into the river • In the river it is oxidised to insoluble iron oxide (orange sediment); the oxidation uses dissolved oxygen so the river is deoxygenated, killing aerobic organisms • Control: drainage water is passed over mesh screens where the iron oxidises and is deposited before the water reaches the river; the solid iron is removed periodically	3
06.4	<i>Any 2 from:</i> • Lead-acid car batteries (recycled) • Lead flashing on roofs / windows • Radiation shielding	2

Question	Marking guidance	Marks
06.5	• They inhibit enzyme function, especially the enzymes of the nervous system / nerve cells	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
07.1	<i>Any 3 from:</i> • Unburnt fuel in vehicle exhaust gases • Evaporation from fuel tanks / spillage at filling stations • Solvents (paints, cleaning products) • Aerosol propellants • Gas leaks / fossil fuel exploitation (methane)	3
07.2	• Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank • Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon	2
07.3	• A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe • Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) • NO_x are chemically reduced to nitrogen and oxygen	3
07.4	• Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues • Brain damage / death if enough haemoglobin is inactivated; low doses worsen chronic heart disease	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	
08.1	<i>Any 4 from:</i> • Sludge from primary sedimentation is digested by anaerobic microbes in a warm tank for about four weeks • Kills most pathogens • Reduces odour and volume of the sludge • Produces biogas (mainly methane with some CO₂) used as fuel to heat the digester / generate electricity • Digested sludge has a better C:N ratio and can be used as agricultural fertiliser	4
08.2	• Heavy metals from urban wastes / road runoff may be present and absorbed by crops - reduced by adding lime to lower metal solubility • Remaining pathogens - destroyed by heat treatment of the sludge before use • Timing: spreading in the growing season covers crops; in winter vehicles compact wet soils - apply at suitable times / when soils are dry • Odour / public objection - digestion reduces odour	4
08.3	• Foul waste is a small volume needing full treatment so smaller, cheaper treatment works can be built; surface water needs little treatment • The treatment works is not affected by weather-related fluctuations in flow (storms) so storm overflows of untreated sewage are avoided	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 270-273; AQA spec 3.4.3.2.9.	
09.1	• B - Steam washing (hot water / steam jets kill organisms in the sediment)	1

Question	Marking guidance	Marks
09.2	- Chemicals cause the oil molecules to join together / solidify - Producing more solid material that can be collected / lifted more easily than liquid oil	2
09.3	- Air contains about 210 cm³ of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm³ per litre (less than 3% of the amount in air) - So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen; many aquatic organisms are near the lower end of their range of tolerance	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
10.1	- No reservoir range = 3.9 - 0.7 = 3.2 m; with reservoir = 2.7 - 1.7 = 1.0 m - Reduction = 2.2 m	2
10.2	- After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream - During dry periods the dam is opened to release water and maintain river levels	2
10.3	- Deforestation / urbanisation reduce infiltration and interception - So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods	2

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Site factors: topography - narrow exit from a large deep basin so a small dam holds a large volume; impermeable, strong bedrock without faults or seismic activity; large catchment with regular high rainfall and low evaporation; catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation; forest / vegetation in the flooded area decaying anaerobically releasing methane; access for workers and materials and distance to demand; value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • Alternatives for supply: rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • Conservation strategies: low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • Evaluation: reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination has high energy / carbon cost; combination needed as demand grows • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

Question	Marking guidance	Marks
11.1	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25

Question	Marking guidance	Marks
11.2	Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Problematic properties of solar: intermittency, unreliability from cloud, low energy density, changing angle; locational constraints (deserts, latitude) • Harnessing solar: photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms; environmental impacts of manufacture and land use • New solar technologies: multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • HEP: gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links; combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	25